A3.1 Clock and reset

This section describes the requirements for implementing the AXI global clock and reset signals **ACLK** and **ARESETn**.

A3.1.1 Clock

Each AXI component uses a single clock signal, **ACLK**. All input signals are sampled on the rising edge of **ACLK**. All output signal changes must occur after the rising edge of **ACLK**.

On master and slave interfaces there must be no combinatorial paths between input and output signals.

A3.1.2 Reset

The AXI protocol uses a single active LOW reset signal, **ARESETn**. The reset signal can be asserted asynchronously, but deassertion must be synchronous with a rising edge of **ACLK**.

During reset the following interface requirements apply:

- a master interface must drive **ARVALID**, **AWVALID**, and **WVALID** LOW
- a slave interface must drive **RVALID** and **BVALID** LOW
- all other signals can be driven to any value.

The earliest point after reset that a master is permitted to begin driving **ARVALID**, **AWVALID**, or **WVALID** HIGH is at a rising **ACLK** edge after **ARESETn** is HIGH. Figure A3-1 shows the earliest point after reset that **ARVALID**, **AWVALID**, or **WVALID**, can be driven HIGH.

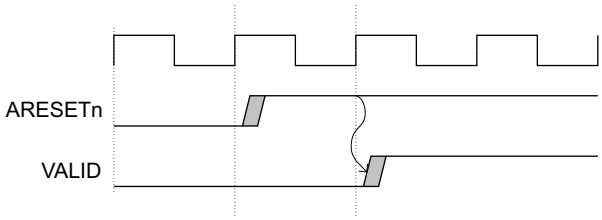


Figure A3-1 Exit from reset

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